Code Modifications:

1. Added #include <semaphore.h> to access POSIX synchronization primitives.
2. Declared ***sem_t driveway_sem***, counts semaphore, and ***mutex counter_mutex***, mutual exclusion lock.
3. Completed ***Cat::run, Cat::wait, Lizard::run***, and ***Lizard::wait***.
4. Added ***sem_wait(&driveway_sem)*** before the lizards and ***sem_post(&driveway_sem)*** after to max the lizards crossing at once to 4 lizards.
5. Global crossing counters of both incremental and decremental operations were wrapped in ***counter_mutex.lock()*** and ***counter_mutex.unlock()*** to prevent simultaneous memory access.
6. **Extra Credit (lizardsUni.cpp):** Added a Lightswitch pattern using an additional binary semaphore to prevent bidirectional travel. The first lizard of a directional group locks the baton, and the last lizard to exit the driveway unlocks it, forcing the opposite lizards to wait.

Table of Results:

WORLDEND (s)	Maximum Number of Lizards Crossing	Lizards safe?
60	2	Yes
120	4	Yes
180	6	Yes
180 (UniDirectional)	4	Yes

Encountered Issues: While testing the code, the main issue was keeping an accurate count of the lizards on the driveway. At first, if multiple lizards tried to cross at the exact same time, the count would get mixed up. Which ended the simulation early. Adding a lock ***(counter_mutex)*** so that only one lizard could change the count at a time fixed this problem. Also, the program would freeze when the timer ran out instead of closing properly. I fixed this by changing the main programs to wait for all the cats and lizards to finish their final actions before shutting down.